Nishanth Panda

bm25mtech14004@iith.ac.in | nishanthpanda@gmail.com | +91 8260351148 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.80
Delhi Public School, Rourkela	Class XII (Central Board of Secondary Education)	2021	81.80%
Delhi Public School, Rourkela	Class X (Central Board of Secondary Education)	2019	96.40%

SKILLS

Technical Skills: Python, C, C++, SQL, MATLAB, PyTorch, TensorFlow, Scikit-learn, Hugging Face, ViT, GenAI/LLMs (RAG, FAISS/ChromaDB, LangChain LoRA, Agentic AI), OpenCV, DICOM Handling, Pandas, NumPy, Matplotlib, Monte Carlo, Node, React, Docker, FastAPI, MLflow, Git, Github, Gradio, Streamlit.

INTERNSHIP EXPERIENCE

Senior Intern (Remote), Shodhaka Life Science Pvt. Ltd., Bengaluru

Skills: Data Scrapping, Data Wrangling, Data Cleaning

- Architected centralized, **researcher-accessible database system** integrating **40+ specialized** biological data sources with automated **ETL pipelines** and **data validation** workflows.
- Implemented **data scraping**, **wrangling**, and quality assurance protocols processing 1000s of records; ensured data integrity across **functional annotations** and **interaction networks**.
- Designed **database schema** and **optimization strategies** for **large-scale** multi-source data aggregation platform.

PROJECTS

Coronary Angiogram Analysis & Segmentation System *(MTech Thesis Project Ongoing)* (github)

***Stack**: Python, HTML5, CSS3, JavaScript, PyTorch, Segment Anything Model (SAM), OpenCV, Scikit-Image, JSON*

- Integrated the **Segment Anything Model (SAM)** with **geometric prompt engineering** and **spatial tensors** to achieve **zero-shot coronary artery segmentation** from **X-ray angiograms**.
- Engineered **Quantitative Coronary Analysis (QCA)** algorithms using **Euclidean distance transforms** and **skeletonization** to automatically calculate stenosis severity and estimate **blood flow reduction**.
- Built a **Python clinical Web GUI** utilizing **SSIM alignment** to visually validate **AI masks** against ground-truth annotations and generate actionable **cardiology metrics**.

Multimodal Chest CT Diagnostics & Explainable AI (github)

***Stack:** PyTorch (ResNet50), PyTorch-Grad-CAM, Scikit-learn, Seaborn, Gradio, Google Gemini API*

- Fine-tuned a **PyTorch ResNet50 CNN** to classify chest CT scans across four oncological categories, achieving **94% test accuracy** and a **0.95 macro F1-score** (including **100% precision/recall** on normal scans and **92-94% recall** on carcinomas).
- Integrated **Grad-CAM (Explainable AI / XAI)** to generate localized **activation heatmaps**, visually highlighting critical anomalous regions on CT slices to make deep learning predictions clinically transparent.
- Deployed the model into an **interactive web dashboard** combining visual inference with an **LLM-based report generator**, delivering real-time, interpretable diagnostic summaries directly from uploaded scans.

Markdown Effect: Second Brain Retrieval System (github)

Tech Stack Used: Python, FastAPI, Uvicorn, Pydantic, ChromaDB, Google GenAI, Python Frontmatter, Hashlib

- Built a scalable **FastAPI** backend with a /sync endpoint to scan **Markdown files**, extract **Frontmatter Metadata**, and compute **MD5 Hashes** for efficient updates.
- Leveraged **Google GenAI** to generate semantic embeddings, seamlessly storing both text and metadata in a high-performance **ChromaDB Vector Database**.
- Implemented a dynamic /search API that combines **Semantic Search** with strict **Metadata Filtering** (by **Tags** and **Date Ranges**) using robust **Pydantic Models**.

DocuMind: AI-Powered PDF Intelligence (github)

Tech Stack: React 19, Vite, Tailwind CSS v4, Axios, Lucide React, Python, FastAPI, LangChain, OpenAI API, Google Gemini API, FAISS

- Engineered a complete **Retrieval-Augmented Generation (RAG)** pipeline using **GPT-4o-mini** & **Gemini 3.5 Flash** for **Context-Grounded Generation**, minimizing **AI Hallucinations**.
- Utilized **LangChain** for effective **Document Segmentation**, and leveraged **OpenAI/Gemini Embeddings** to store vector data in a highly efficient **FAISS Vector Database**.
- Designed a responsive **React/Vite** frontend and **FastAPI** backend that map retrieved answers to original **Metadata (Page Numbers & Snippets)** for transparent **Citations**.

CERTIFICATIONS

Digital Skills: Artificial Intelligence - Accenture & FutureLearn (CPD Certified):

Completed coursework covering AI and Generative AI fundamentals, industrial automation, and human-AI collaboration, achieving a 75% score.

POSITIONS OF RESPONSIBILITY

Placement Coordinator - Office of Career Services (OCS), IIT Hyderabad.

Campus Ambassador, Entrepreneurship Cell, IIT Hyderabad

- Supported the coordination of entrepreneurial events and student-led initiatives to foster startup innovation among youth.